

Agile Backlog Creation & Sprint Simulation

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Problem statement 16: Remote Patient Vitals Alert & Monitoring App. The system receives SpO2, heart rate and blood pressure, detects configured threshold breaches and alerts caregivers through an escalation matrix.

Individual scope: RPM-3 - Emergency Alerts and Caregiver Escalation (main focus), and RPM-4 - Secure Caregiver Dashboard and History (additional scope). Together these cover caregiver notification, acknowledgement, escalation, authorized monitoring and history review.

Epic	Scope / Lab 1 traceability
RPM-1 Patient Telemetry Ingestion	FR-001: receive, validate and store telemetry; recovery is a backlog refinement
RPM-2 Clinical Threshold and Anomaly Detection	FR-002: configure limits and detect critical anomalies
RPM-3 Emergency Alerts and Caregiver Escalation	FR-003/FR-004: notification and escalation; FR-005: acknowledgement
RPM-4 Secure Caregiver Dashboard and History	FR-005: authorized access, current vitals and alert history

Jira setup: Company-managed Scrum project RPM, board 2; four epics and twelve user stories. Stories use the As a / I want / So that format, acceptance criteria, priorities and Fibonacci estimates.

Exercise method: Each sprint was configured for seven days. To Do - In Progress - Done movements were simulated in a shortened exercise, as permitted by the lab handout. Done denotes simulated workflow completion, not implemented or clinically validated software.

RPM-3: Alerts & Escalation

RPM-11: Deliver critical alerts to the on-call caregiver | High priority | 5 story points

As an on-call caregiver, I want critical alerts with patient and vital details, so that I can respond promptly.

Acceptance criteria: Route to the first eligible caregiver; include patient, vital, value, threshold and time; deliver at least 95% of critical notifications within 5 seconds after detection in a representative test; log failures.

Estimate rationale: Notification delivery, routing and latency testing.

RPM-12: Acknowledge an alert and record responsibility | High priority | 3 story points

As an on-call caregiver, I want to acknowledge an alert, so that the system records who is responding and stops further escalation.

Acceptance criteria: Record authorized caregiver identity and acknowledgement time; set Acknowledged; cancel pending escalation; repeated requests are idempotent. Acknowledgement does not imply clinical resolution.

Estimate rationale: Bounded state change, authorization and timer cancellation.

RPM-13: Escalate unacknowledged alerts through the caregiver matrix | High priority | 8 story points

As a remote patient, I want unanswered critical alerts escalated to backup caregivers, so that another caregiver can respond.

Acceptance criteria: After a configured timeout, notify the next eligible caregiver in order; log each escalation; stop after acknowledgement; handle acknowledgement/timeout races; record matrix exhaustion.

Estimate rationale: Timers, retries and concurrent acknowledgement create greater uncertainty.

Epic total: **16 points**, included in Sprint 2 alongside the 3-point dashboard story RPM-15. Estimates are provisional; no completed team Planning Poker session is claimed.

RPM-4: Secure Caregiver Dashboard & History

Lab 1 link: FR-005 covers authorized caregiver access, current vitals and alert history. RPM-12 in the alert epic handles acknowledgement from the dashboard.

RPM-14: Restrict dashboard access by role and patient assignment | High priority | 5 story points

As a remote patient, I want only authorized users to access my monitoring data, so that my health information remains private.

Acceptance criteria: Require authentication for vitals and history; allow caregivers access only to assigned patients and patients access only to their own data; deny attempts using changed patient IDs; log authorization failures without secrets.

Estimate rationale: Authentication integration and patient-level authorization tests. **Status:** Done in Sprint 1 (simulation).

RPM-15: Display current vitals and active alerts | High priority | 3 story points

As an on-call caregiver, I want to see assigned patients' latest vitals and active alerts, so that I can identify patients needing attention.

Acceptance criteria: Show SpO2, heart rate, systolic/diastolic BP, units and measurement time; highlight critical alerts and acknowledgement state; label stale or disconnected data; refresh when newer authorized readings arrive.

Estimate rationale: Dashboard presentation and integration with telemetry updates. **Status:** Done in Sprint 2 (simulation).

RPM-16: Review and filter patient alert history | Medium priority | 3 story points

As an on-call caregiver, I want to filter a patient's alert and acknowledgement history, so that I can review events during follow-up.

Acceptance criteria: Filter authorized history by patient, date range and alert state; show detection, delivery, escalation and acknowledgement times where present; display an empty-state message when no results match; enforce the same patient access restrictions as live data.

Estimate rationale: History queries and filters using established authorization. **Status:** To Do; retained in the backlog, outside both completed sprints.

Scope totals: RPM-4 has 11 points: 8 simulated Done and 3 in the backlog. With RPM-3, this report covers 27 points: 24 simulated Done and 3 pending. Estimates remain provisional pending team review. Patient self-access in RPM-14 is a backlog elaboration beyond Lab 1's caregiver dashboard requirement.

Sprint Results & Reflection

Sprint	Planned scope	Simulated completion
Sprint 1	RPM-5, RPM-6, RPM-14, RPM-8, RPM-9 21 points	5/5 stories Done; 21/21 points
Sprint 2	RPM-11, RPM-12, RPM-13, RPM-15 19 points	4/4 stories Done; 19/19 points

Sprint 1 planned: 2 Oct 2026, 20:13 to 9 Oct, 20:13 IST; closed 3 Oct, 01:09 IST.

Sprint 2 planned: 3 Oct 2026, 01:09 to 10 Oct, 01:09 IST; closed 3 Oct, 01:14 IST.

Three medium-priority stories (RPM-7, RPM-10, RPM-16; 11 points) remain in the backlog.

1. Did the estimations reflect actual effort?

The estimates represent relative implementation complexity, not measured development time. Escalation received 8 points because of timers, retries and race conditions; acknowledgement received 3. Since this was a workflow simulation, actual implementation effort and estimation accuracy could not be validated.

2. Was the backlog well-prioritized?

High-priority work covered reliable ingestion, access control, detection and emergency response. Supporting recovery, incident grouping and history filtering remained in the backlog. Sprint 1 established foundations; Sprint 2 covered notification, acknowledgement, escalation and the dashboard.

3. How did the simulated sprint align with the plan?

Both sprints completed their selected scope without adding or removing stories during the sprint. All nine selected stories passed through In Progress before Done. The planned timebox was one week, while the exercise deliberately compressed workflow progress.

4. What did the burndown chart show about team capacity?

The remaining work fell from 21 to 0 and from 19 to 0. Sharp early drops reflect the shortened simulation, not sustained weekly productivity. The completed points demonstrate Jira tracking; they cannot establish real team capacity. Future delivery forecasts should use several actual development sprints.

RPM-4: Child Stories, Estimates & Status

For you

Recent

Starred

Apps

Plans

Spaces

Recent

Remote Patient Monitoring

Recent

RPM board

ABC

More spaces

Recommended

Prioritize ideas

Filters

Dashboards

Assets

Teams

Goals

Projects

Customise sidebar

Search

Create

See plans

3+

VA

Spaces / Remote Patient Monit... / RPM-4

Backlog for the Lab 2 sprint simulation. Workflow completion in this exercise represents simulated progress, not a deployed clinical system. Team member ownership is pending; each member should select a distinct epic for the individual report.

Attachments

Child work items

Linked work items

Activity

1

To Do

Improve Epic

Details

Assignee

Unassigned

Assign to me

Reporter

Vivaan Atharva

Labels

ps16 se-lab2 sprint-simulation

Due date

Add due date

Start date

Add date

Priority

High

Development

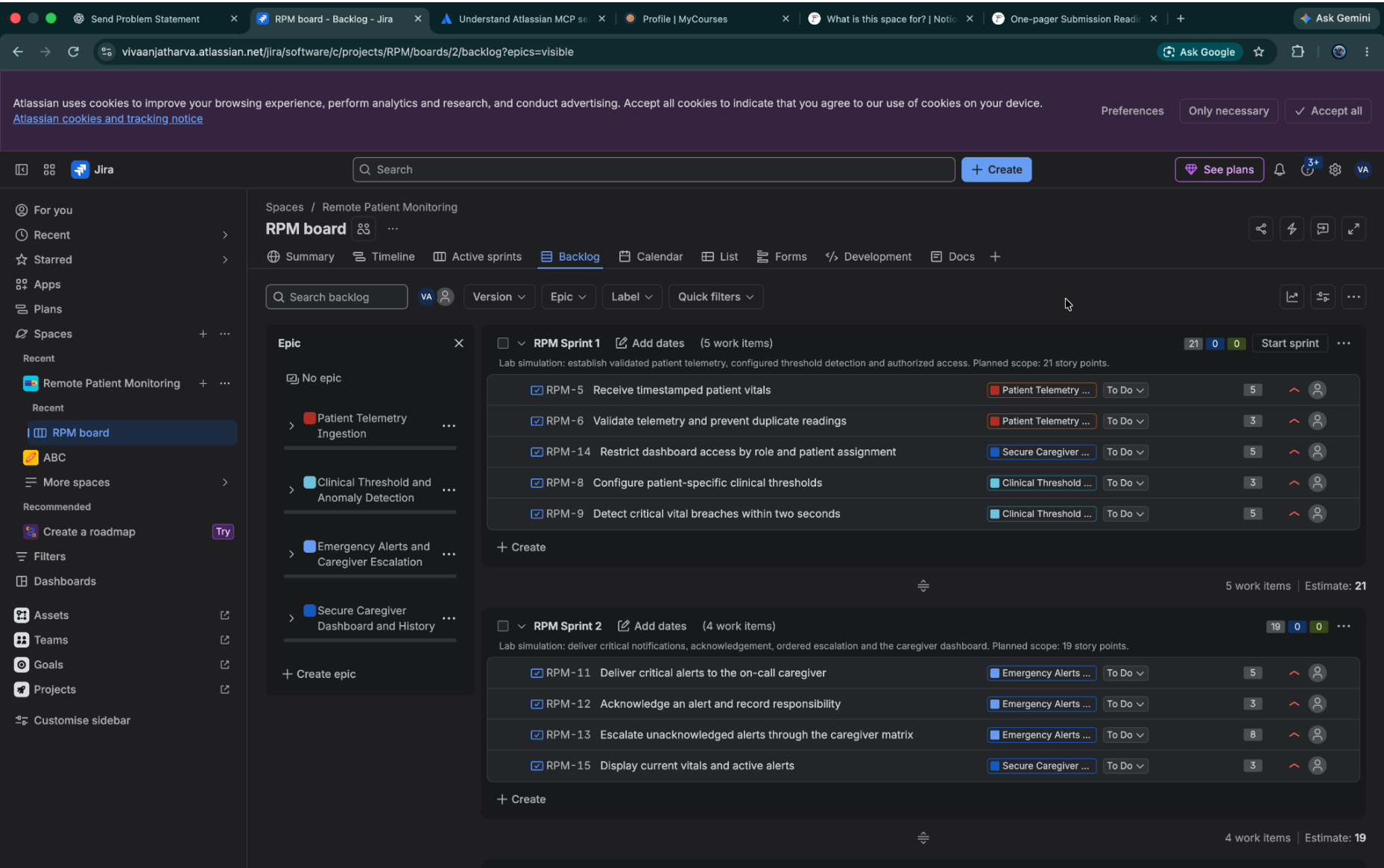
Work	Pri...	Stor...	As...	Status	
☒ RPM-14	Restrict dashboard access by role and patient assignment	H.	5	U	Done
☒ RPM-15	Display current vitals and active alerts	H.	3	U	Done
☒ RPM-16	Review and filter patient alert history	M	3	U	To Do

RPM-14: 5 points, Done; RPM-15: 3 points, Done; RPM-16: 3 points, To Do. Two of three stories are Done (8 of 11 points).

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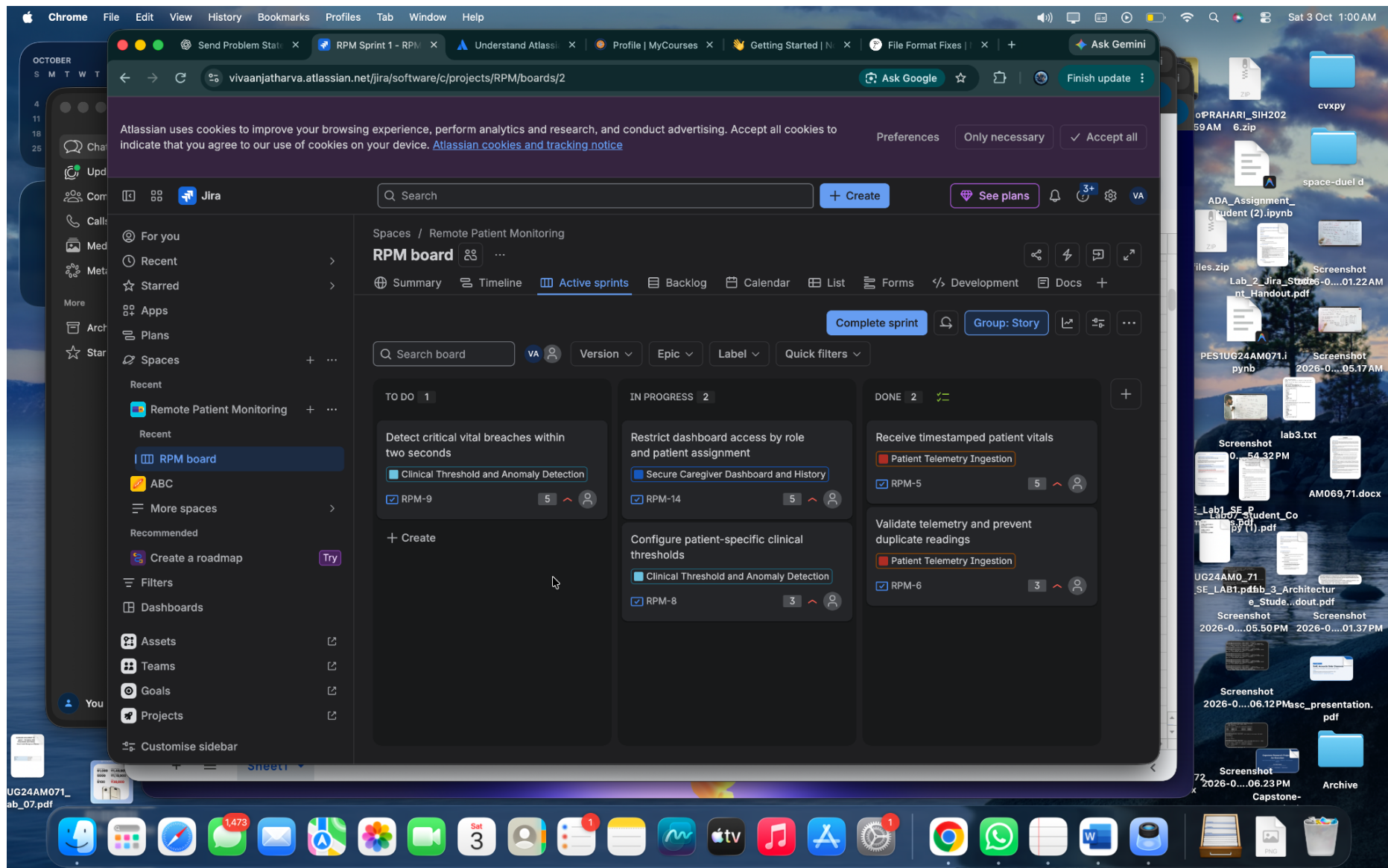
5

Backlog, Epics & Story Points



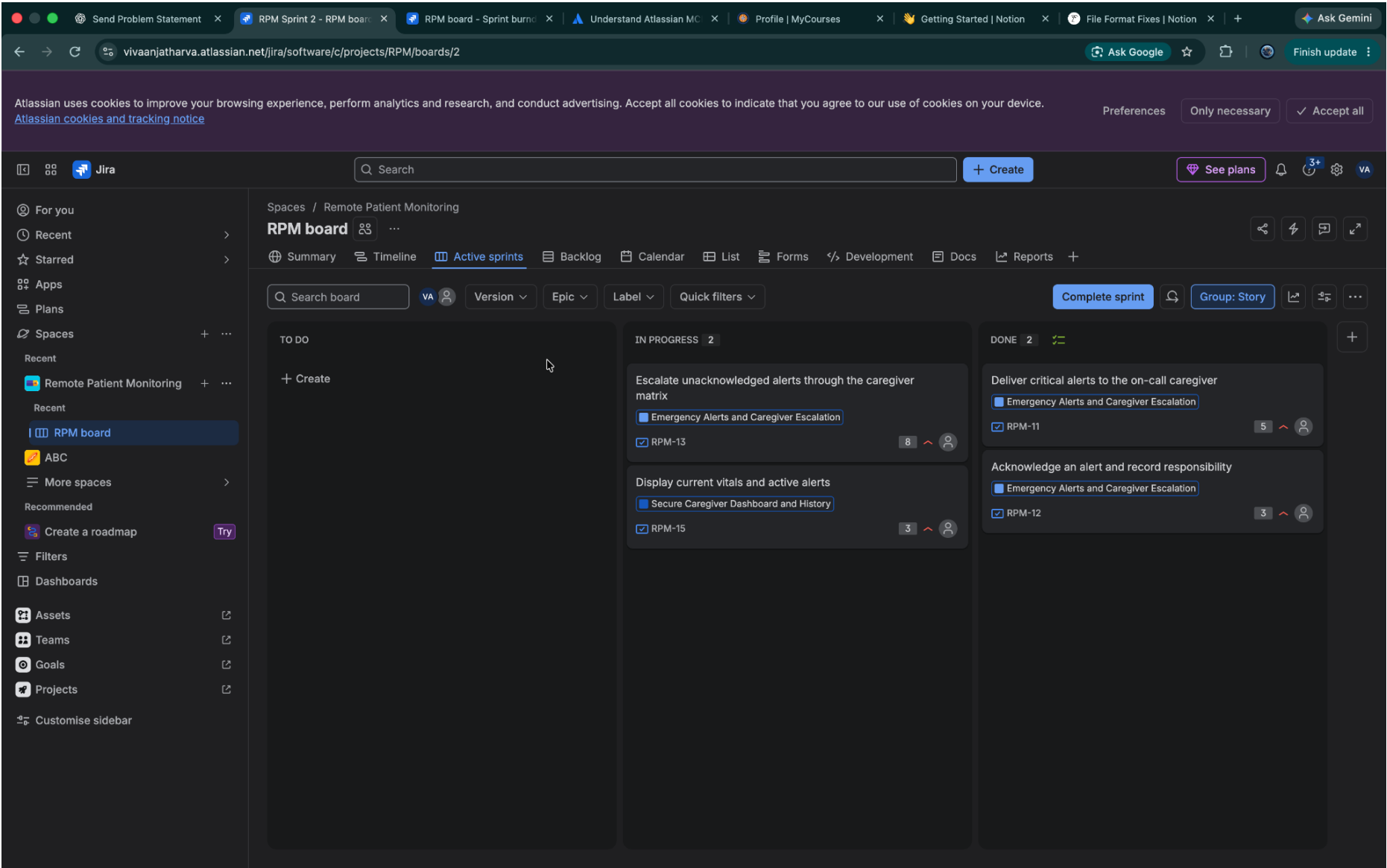
Four epics and both planned sprints. Story estimates and priorities are visible before sprint execution.

Sprint 1: Active Board



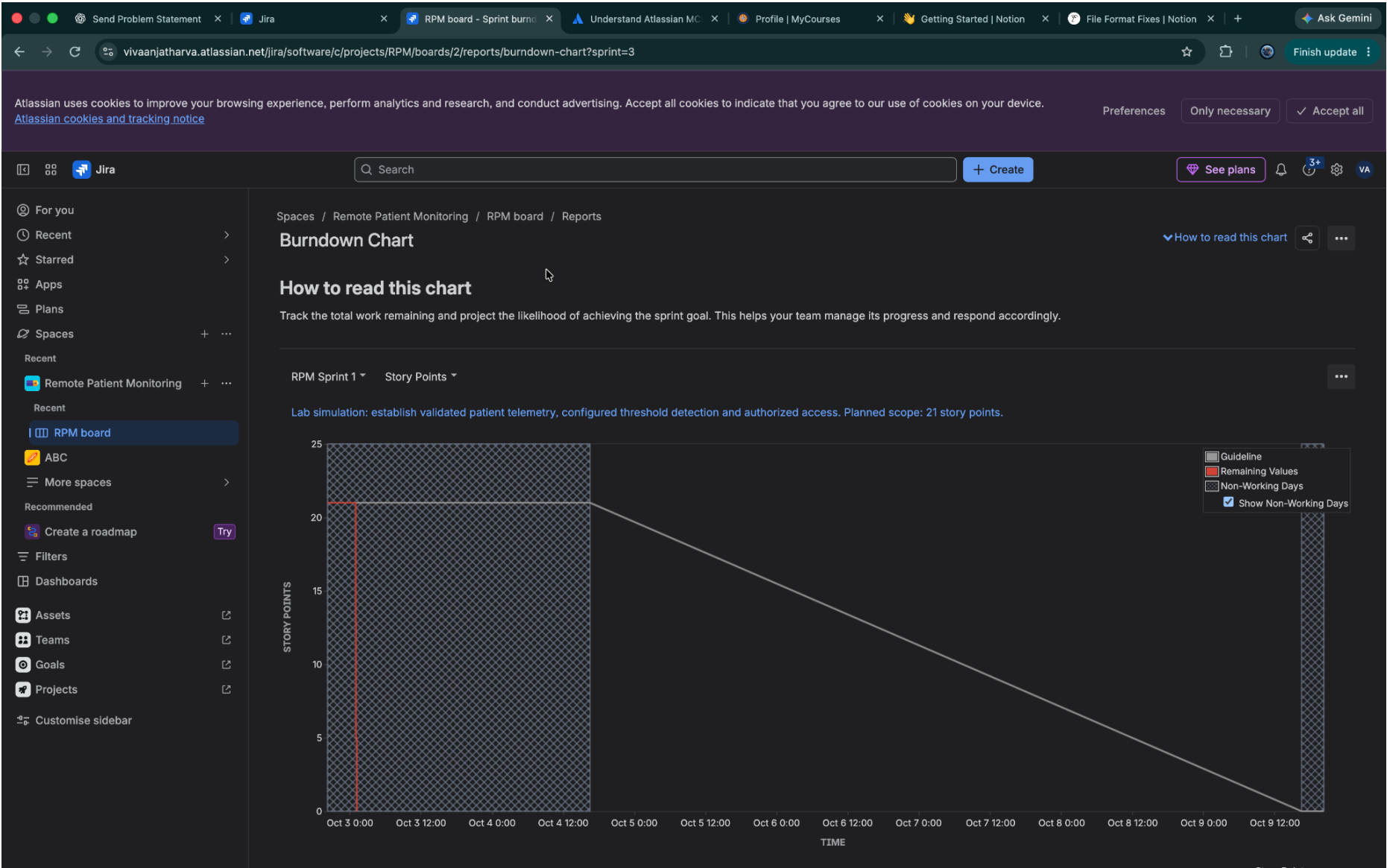
Intermediate state: 2 Done (8 points), 2 In Progress (8 points), 1 To Do (5 points).

Sprint 2: Active Board



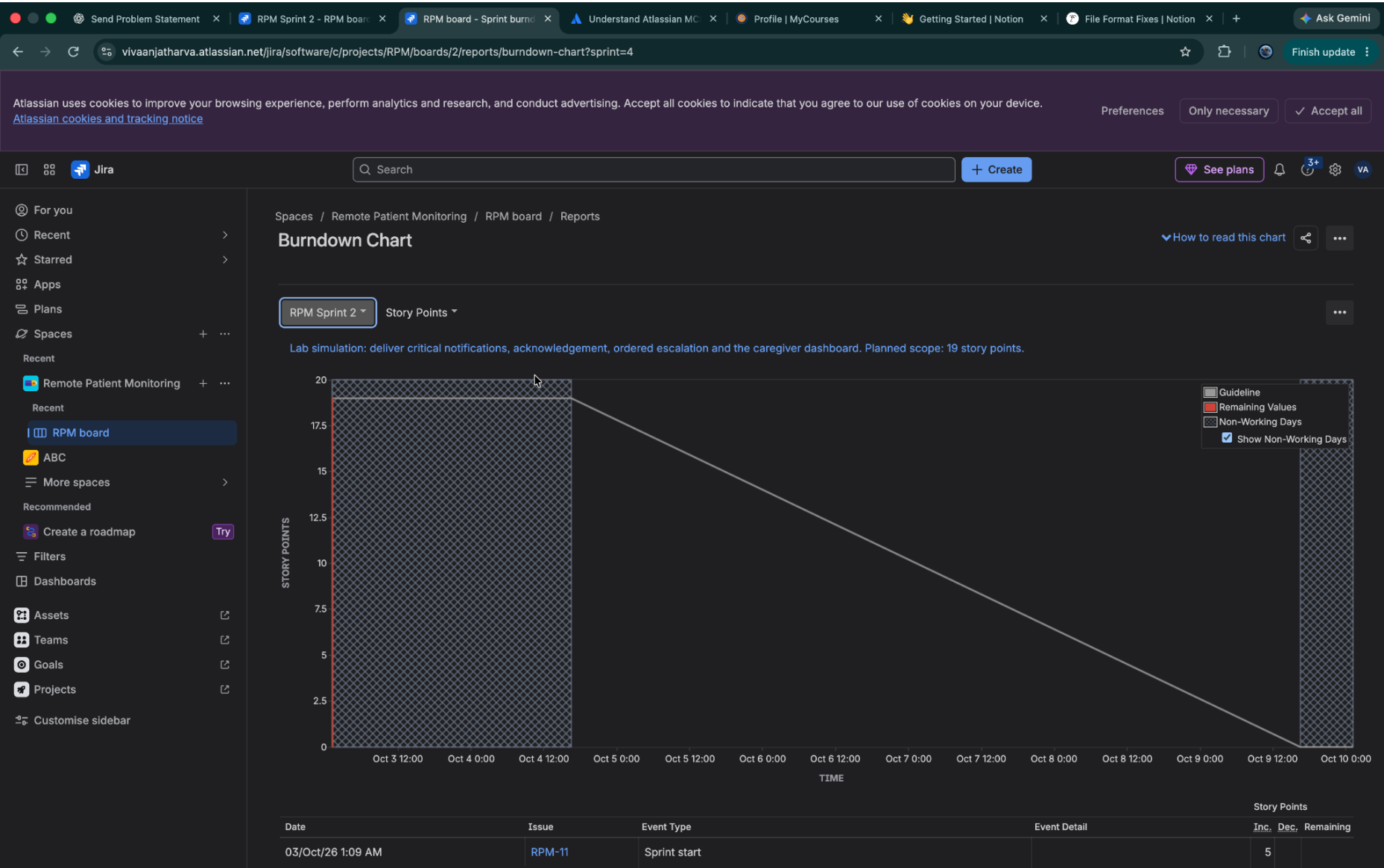
Intermediate state: 2 Done (8 points), 2 In Progress (11 points).

Sprint 1: Final Burndown



Final remaining work: 0 of 21 points. The early drop reflects the shortened lab simulation.

Sprint 2: Final Burndown



Final remaining work: 0 of 19 points. The one-week guideline and actual simulated completion are shown.